

Problem D. Points On a Line

OS Linux

Given n two-dimensional points in space, determine whether they lie on some vertical or horizontal line. If yes, print YES; otherwise, print NO.

Input Format

The first line contains a single positive integer, n , denoting the number of points. Each line i of n subsequent lines contain two space-separated integers detailing the respective values of x_i and y_i (i.e., the coordinates of the i^{th} point).

Constraints

- $2 \leq n \leq 10$
- $-10 \leq x_i, y_i \leq 10$

Output Format

Print YES if all points lie on some horizontal or vertical line; otherwise, print NO.

Input	Output
5 0 1 0 2 0 3 0 4 0 5	YES

Explanation 0

All points lie on a vertical line.

Input	Output
5 0 1 0 2 1 3 0 4 0 5	NO

Explanation 1

The points do not all form a horizontal or vertical line.